26900113 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

Regular EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECHEA03 - MANUFACTURE OF PETROCHEMICALS II

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Methanol is industrially manufactured from synthesis gas, which mainly consists of: [U] / [CO1]

   a). CH_4 and O_2 b). CO and H_2

   c). CO_2 and O_2 d). CH_4 and CO_2

2. Which property of carbon black makes it useful as a reinforcing agent in rubber? [U] / [CO1]

   a). High surface area b). High solubility in water

   c). Low thermal stability d). High volatility

3. Why are the product gases cooled in the Andrussow process? [U] / [CO1]

   a). To prevent formation of azulumic acid

   b). To increase ammonia formation

c). To remove methane

d). To increase the boiling point of HCN

4. Which chloromethane is widely used as a paint remover? [U] / [CO2]

a). CH_3Cl b). CH_2Cl_2 c). $CHCl_3$ d). CCl_4

5. Which factor contributes most to the clean-burning nature of DME? [An] / [CO2]

a). High sulphur content

b). Oxygen in the molecule

c). Aromatic structure

d). High viscosity

6. Select the product formed by aldol condensation of n-butyraldehyde. [U] / [CO3]

a). Acetone b). 2-Ethylhexenal

c). Acrylic acid d). Allyl alcohol

7. Identify the aldehyde hydrogenated to form iso-butanol. [K] / [CO3]

a). n-Butyraldehyde b). Iso-butyraldehyde

c). Acrolein d). Acrylic acid

8. Methanol is historically known as: [U] / [CO1]

a). Grain alcohol b). Wood alcohol

c). Rubbing alcohol d). Fuel alcohol

9. Acrylonitrile is also known as_____ [K] / [CO4]

a). Vinyl cyanide b). Vinyl alcohol

c). Vinyl acetate d). Vinyl ether

10. Select the process which has largely replaced the acetylene route for vinyl chloride production? [U] / [CO4]

a). Balanced ethylene process b). Reppe process

c). Wacker process d). Sohio process

[1x10 = 10]

Part B

Short Answer Questions

Answer any **8** questions

Each question carries **3** marks

11.	Describe the Furnace Process for the manufacture of carbon black.	[U] / [CO1]
12.	State the industrial uses of carbon tetrachloride (CCl_4).	[K] / [CO2]
13.	List the important applications of formaldehyde.	[K] / [CO2]
14.	Describe the Shawnigan process for the manufacture of HCN.	[U] / [CO1]
15.	A rubber industry requires a chemical for preparing rubber accelerators. Which methane-derived product would you recommend? Justify.	[A] / [CO2]
16.	Name the catalysts used in the hydroformylation of propylene.	[K] / [CO3]
17.	Explain the formation of glycerin during soap manufacture.	[U] / [CO3]
18.	Explain the function of a quench tower in acrylic acid manufacture.	[U] / [CO3]
19.	Outline why acrylonitrile is considered an important industrial monomer?	[An] / [CO4]
20.	Acetylene is called a versatile petrochemical feedstock. Relate with examples.	[U] / [CO4]

[3x8 = 24]

Part C

Short Essay Questions

Answer any **3** questions

Each question carries **7** marks

21.	Discuss the thermal process for the manufacture of carbon black. Describe its characteristics and important applications.	[U] / [CO1]
22.	Explain the production of formic acid from methanol and carbon monoxide through methyl formate.	[U] / [CO2]
23.	Describe the manufacture of 2-ethylhexyl acrylate from propylene.	[U] / [CO3]
24.	Explain the principle and process of acetylene manufacture by the Sachse process.	[U] / [CO2]
25.	Explain Reppe synthesis of acrylic acid.	[U] / [CO4]

[7x3 = 21]

Part D

Essay Questions

Answer any **1** questions

Each question carries **15** marks

26. Discuss the manufacture of isopropanol from propylene by the direct hydration method. Explain the reaction, process steps, separation, purification and uses of isopropanol. [U] / [CO3]

27. Explain the manufacture of acetaldehyde from acetylene with a neat process description and operating conditions. [U] / [CO4]

[15x1 = 15]